

Paul Abrams

Toronto, Ontario

Senior Software Engineer — C# / .NET · Microservices · High-Scale Backend

(819) 921-7192 | j.paul.abrams@protonmail.com | github.com/pabrams | paulabrams.ca

Professional Summary

Senior full-stack .NET developer with 15+ years shipping enterprise web applications, now building AI-integrated systems hands-on: a Model Context Protocol (MCP) tool-server with in-code agent guardrails, daily LLM-assisted development with Claude Code, and professional evaluation of LLM-generated code via the Anthropic API.

Professional Experience

Abrams Software Toronto, Ontario

August 2025 – Present

Full-Stack Developer · Independent

- Built CVForge, a blurb-library and CV-builder product — a Vue.js 3 single-page front end over an ASP.NET Core (C#) REST Web API with relational SQL persistence.
- Designed the REST API endpoints consumed by the SPA client and by an MCP tool-server integration.
- Shipped other production web applications solo, including a multiplayer game (Node.js, PostgreSQL, with a REST API contract generated from JSON Schema) and a React 19 + TypeScript chess visualization consuming the Lichess REST API.

Abrams Software Toronto, Ontario

August 2025 – Present

LLM / AI Code & Security Evaluation

- Authored and QC-reviewed cybersecurity capability-evaluation tasks that measure whether AI coding agents can identify and exploit real software vulnerabilities. For each task, reproduced a published advisory (GitHub Security Advisory / Go vuln DB) as paired vulnerable and patched Docker Compose environments, wrote a reference exploit proof-of-concept, and built an automated differential grader that verifies exploitation through observable side-effects (out-of-band callbacks, canary tokens) — passing on the vulnerable build, failing on the patched one, and hardened against grader-gaming.
- Covered deserialization RCE, SSRF, path traversal, authentication bypass, cryptographic, logic, and race-condition vulnerability classes across Java, Go, Python, TypeScript/Node, C++, and Ruby target codebases. Calibrated task difficulty by running a reference model one-shot against a target pass rate using the Anthropic API, and ran dual-LLM-judge QC review scoring vulnerability match, prompt leakage, and grader soundness.
- Broader LLM code evaluation for DataAnnotation.tech and Stellar.ai — prompt fixtures and grading rubrics across .NET/C#, TypeScript, React, Node/NestJS, Python, Go, Rust, SQL, and MongoDB, including CWE/CVSS-graded security audits.

Abrams Software Toronto, Ontario

August 2025 – Present

Full-Stack Developer · Independent

- **TWNR — architected and shipped a public-facing B2C multiplayer game, solo.** Dockerized **TypeScript** monorepo: **Node.js (Express) + WebSocket** server backed by **PostgreSQL**, per-player **JWT auth**, and a **JSON-Schema-generated REST API contract** that doubles as live documentation. Concurrency-correct state via database transactions. Live at twnr.paulabrams.ca.

Employment and Social Development, Canada Gatineau, QC

December 2021 – July 2025

Senior Software Developer — Cloud-Native Microservices

- Enhanced and maintained a cloud-native microservices onboarding application (.NET Core / C#, Blazor, Azure SQL, Azure Functions).
- Designed and consumed REST APIs between services, documented with Swagger and governed through Azure API Management.

- Migrated downstream identity to Azure Managed Identities, eliminating the secret-rotation incident class.

Employment and Social Development, Canada Gatineau, QC August 2019 – November 2021

Senior Software Developer — Internal Web Application & CI/CD Automation

- Co-architected and developed an internal C# / .NET web application — web UI for authoring structured automation scenarios, SQL Server relational schema design, and a co-designed REST API providing bidirectional integration with Azure DevOps pipelines.
- Engineered Azure DevOps pipelines that provisioned Windows agents to run parallelized automated runs against deployed applications; enforced coding standards through code review.
- Drove a tooling consolidation that cut software licensing costs while preserving capability.

Department of Immigration, Refugees, and Citizenship, Canada

Ottawa, ON

August 2017 – August 2019

Team Lead / Technical Lead — Platform & Tooling

- Technical leadership of a cross-functional team — led Agile / Scrum sprint planning, review, retrospective, daily stand-up, and backlog refinement; introduced poker-planning estimation and capacity forecasting; set engineering standards.
- Mentored developers in SQL Server query optimization — query tuning, indexing strategy, partitioned tables, and execution-plan analysis — and coached on C# / .NET.
- Administered departmental Azure DevOps Server / TFS instances, maintained heavily customized process templates, and managed user/group security across multiple project collections.

Department of Justice, Canada Ottawa, ON

December 2015 – August 2017

Acting Lead Developer — iCase / Microsoft Dynamics CRM Modernization

- Lead developer on the live Legal Case Management System (iCase) — a large n-tier ASP.NET / C# / ADO.NET enterprise web application used by 5,000+ government lawyers — while concurrently building its Microsoft Dynamics CRM successor.
- Engineered Dynamics CRM plug-ins, custom actions, and workflows in C# (rewriting legacy stored procedures as C# / LINQ); used SSIS for ETL; produced entity-relationship diagrams and multi-month effort-estimation breakdowns (FDD / TDD).
- Established code-review processes gated on work items and mentored junior developers.

Department of Justice, Canada Ottawa, ON

October 2008 – April 2014

Full-Stack Developer — Legal Case Management System (iCase)

- Core developer on a distributed, multi-tier ASP.NET / C# enterprise platform used by 5,000+ government lawyers nationally.
- Optimized a high-traffic backend report for a ~50x speedup via execution-plan analysis.
- Migrated core business modules from VB.NET to C#.

Department of Justice, Canada Ottawa, ON

October 2008 – April 2014

Software Development Engineer in Test / Full-Stack Developer — iCase

- Architected a **Java-based automation framework** that **parsed English-language scenario documents into executable scripts with round-trip code generation** — replacing a two-week, ~20–25-person manual pre-release validation cycle with automated execution reviewed by one person (**~98% reduction in effort, ~\$160K saved per release cycle**).

Education

Bachelor of Computer Science (Honours) - Carleton University

September 1999 – May 2003

Technical Skills

Back-End (.NET): C#, ASP.NET (Web Forms → Core), ASP.NET Web API / REST, .NET Framework 2.0 – 4.8 and .NET 5 – 8, VB.NET (legacy migration to C#), Entity Framework, LINQ, ADO.NET, XML / XSLT, WCF / WPF, IIS, Visual Studio 2005 – 2022.

Databases: MS SQL Server 2005 → 2019, Azure SQL, PostgreSQL — relational modelling, T-SQL stored procedures, indexing, partitioning, execution-plan tuning.

APIs: RESTful Web API design between services — documented and tested with Swagger (OpenAPI) and Postman; published and governed through Azure API Management; SOAP / WCF on legacy tiers.

AI / LLM Tooling: Model Context Protocol (MCP) tool-server design with in-code agent guardrails · Claude Code (daily development driver) · professional LLM code evaluation (prompt fixtures, grading rubrics) · AI cybersecurity capability-evaluation task authoring (containerized vulnerable/patched targets, differential automated graders, difficulty calibration via the Anthropic API).

Cloud & Containers: Microsoft Azure — App Service, Azure Functions, AKS, API Management, Key Vault, Managed Identities, Azure Monitor / Application Insights / Log Analytics · Docker containerization and Kubernetes / AKS deployment · Red Hat OpenShift (formal training + production support tickets) · Terraform infrastructure as code · GraphQL alongside REST · event-driven async messaging (queue and storage triggered Azure Functions) · Azure DevOps Pipelines (YAML), GitHub Actions.

DevOps & Process: Azure DevOps Services and Azure DevOps Server / TFS (all supported versions — installation, upgrades, administration), Git, CI/CD pipelines · Agile / Scrum (Professional Scrum Master I) and waterfall delivery · code review, mentoring, JAD sessions, FDD / TDD documentation and effort estimation.

Security: OAuth 2.0 / OpenID Connect / PKCE, JWT, Entra ID / Azure AD IAM, Managed Identities, Key Vault secret hygiene, TLS / certificate management, CORS policy, SonarQube (SAST), OWASP ZAP (DAST), CWE / CVSS-graded security audits, NIST-based (ITSG-33 / PBMM) controls, secret-scanning guardrails. Professional Scrum Master I (Scrum.org) · ITIL v3 Foundation (Loyalist Certification Services).

Open source (60+ public PRs): chess.js #572 — x-ray attacker detection, with a published fork on npm as @pabrams/chess.js · Microsoft vsts-team-calendar #207 — es6-promise polyfill fixing “Promise is not defined” in IE · azure-devops-node-api #159 · azure-devops-extension-hot-reload-and-debug #3 · Microsoft TypeScript-Website #3477 / #3478 · microsoft-authentication-library-for-dotnet #317 — PowerShell support docs · physicshub (three fixes) · canada-ca/platform-security · Dygmalab/Bazecor (Dyigma keyboard configurator).

Awards (Department of Justice, Canada): Team Merit Award 2012 (Cost Recovery Process Improvement) · Team Merit Award 2011 (Chart of Accounts) · Team Spirit Award 2009.

Core Competencies

Senior software developer with 15+ years building, scaling, and operating high-availability, distributed enterprise systems across the Government of Canada — from a **load-balanced, geographically distributed** legal case-management platform serving **5,000+ concurrent users nationally** to a **cloud-native microservices** application on Microsoft Azure. Deep **backend** expertise in **.NET Core / C#** and **Node.js / NestJS** (a TypeScript decorator framework analogous to **Spring Boot** / ASP.NET Core), designing and consuming **REST APIs between services** and publishing/governing them through an **API gateway (Azure API Management)** with authentication, throttling, and transformation. Hands-on with **Docker and Kubernetes** containerized deployment, **CI/CD pipelines** (Azure DevOps, GitHub Actions), **serverless / event-driven workflows** (Azure Functions, queue- and storage-triggered handoffs), and **load balancing** across cloud and on-prem tiers. Strong relational-database engineering (**MS SQL Server, Azure SQL, PostgreSQL**) with working **NoSQL (MongoDB)** exposure, and a track record of **optimizing high-traffic backends** (e.g. a ~50× report speedup via execution-plan analysis). **Security-minded by default:** OAuth 2.0 / OpenID Connect / Entra IAM, migration to **Managed Identities, Key Vault** secret hygiene, TLS / certificate management, and **ITSG-33 / PBMM** controls. **Recurring pattern:** self-initiated automation frameworks that pay back substantial labour cost. **Professional Scrum Master I; AODA / WCAG-aware.** Active open-source contributor — **60+ public PRs** including Microsoft’s vsts-team-calendar, tfs-cli, and the **Microsoft Authentication Library .NET** docs.